

Intro to R

Day 1

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Scripts, buttons, running code, and the console

RStudio and Positron

- R vs the IDE
 - R: language and interpreter
 - IDE: interface for writing, running, and inspecting code
- Panes
 - Source editor: write and save scripts
 - Console: send code to R and see results
 - Environment: view session objects
 - Files, plots, packages, help: supporting views
- Main controls
 - New, open, save script
 - Runing current line or a selection
 - Source entire script
 - Stopping running code
 - Restarting R

Script versus console

- Script
 - Saved source of the analysis
 - Can be run in full or in part
 - Comments and section breaks
- Console
 - Immediate experiments
 - Included in history
- Dangers of the consule
 - Edit the environment but aren't documented
 - Can break reproducibility

Running code

- Things that can be run
 - Line
 - Selection
 - Whole script
- Top-to-bottom execution
- Feedback from the code
 - Printed output
 - Messages and warnings
 - Errors

Environment and why not to save it

The environment as session state

- Named objects in memory
- Environment pane
 - Object inventory
 - Current values
- Session lifetime
 - Objects created when running old code
 - Objects removed when R restarts

The hidden-state problem

- Stale objects from earlier work
- Objects created out of order
- Different results across machines or sessions

Saved workspace versus reproducible analysis

- Saved workspace
 - Creates an .RData
- Objects reappear without rerunning code
- Costs
 - Hidden inputs
 - Unclear history of the object
- Preferred workflow
 - Write and edit code in a script
 - Save any necessary outputs explicitly
 - Once ready, run the script in a fresh environment

Packages, libraries, and variables

Packages: reusable code

- How we share code

- All R code that you see in Base R is from packages
- What a package contains
 - Functions
 - Documentation (vignettes, examples)
 - Example data
- Installation
 - Once per R install-package version pair
 - e.g. `install.packages("readr")`
- “Base R” vs. “contributed packages”

Libraries: packages in the current session

- Loading a package
 - Once per session
 - Example: `library(readr)`
- Explicit namespace
 - Example: `readr::read_csv()`
- Flag for the future
 - May see `devtools::load_all()`; this loads a local library

Variables: named objects

- Object as a named value
- Assignment
 - Name gets value (`x <- 1`)
 - Inspecting the value by printing (`x`)
 - Reassignment by assigning after (`x <- 2`)
- Naming conventions
 - Descriptive names (don't use `x`)
 - `snake_case`
 - Case sensitivity
 - No spaces

Working with variables

- Variables as a superset: anything can be variable
- Preview: types
 - `str(x)`
 - `class(x)`
- `length(x)` and `lengths(x)`